

SIMARPREET SINGH

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EDUCATION

I.K. Gujral Punjab Technical University, Kapurthala

2022 - 2026

B. Tech in Computer Science and Engineering

9.00 CGPA

EXPERIENCE

CDAC (Centre for Development of Advanced Computing)

Mohali

Artificial Intelligence & Computer Vision Intern

January 2026 – July 2026

- Completed a 6-month internship focused on Artificial Intelligence and Computer Vision using Python.
- Implemented data preprocessing and analysis using Python, NumPy, Pandas, and Matplotlib.
- Applied OpenCV techniques for image processing, filtering, and feature extraction tasks.
- Gained practical exposure to supervised and unsupervised machine learning concepts.
- Worked on dataset preparation, model evaluation, and performance assessment workflows.

Ryazio Technologies LLP

Remote

Industrial Training

June 2024 – July 2024

- Built and optimized **full-stack** applications, focusing on performance and scalability.
- Enhanced skills in **RESTful API** development, authentication, and database management.
- Worked on **real-world** client projects, improving code quality and user experience.
- Strengthened problem-solving and industry-specific development practices.

Novem Controls

Remote

Web Development Internship

June 2024 – July 2024

- Developed and maintained full-stack web applications using the **MERN** stack.
- Designed responsive, user-friendly interfaces with **React.js** and **Tailwind CSS**.
- Implemented backend **APIs** with **Node.js** and **Express**, integrating **MongoDB** for scalable data storage. Collaborated with team members to deploy projects and resolve production-level issues.
- Gained hands-on experience in live project development and deployment workflows.

PROJECTS

Advanced AI Interview System | *Python, FastAPI, SQLAlchemy, JavaScript*

- Developed an AI-powered interview platform featuring resume parsing, automated question generation, and real-time candidate evaluation.
- Integrated Whisper-based speech transcription, facial emotion analysis, and video recording to provide comprehensive interview insights.
- Implemented integrity monitoring mechanisms, including tab-switch detection, face verification, and audio-based anomaly checks to ensure assessment fairness.
- Built an interactive admin dashboard for managing candidates, reviewing transcripts, analyzing performance scores, and accessing recorded interview sessions.

BRSLMS – Full-Stack Learning Management System | *MERN Stack*

- Built a full-stack Learning Management System using React, Node.js, and Express to manage students, teachers, and academic data.
- Developed RESTful APIs for attendance tracking, assignments, and academic performance management.
- Implemented role-based authentication and authorization for admin and faculty users.
- Created an admin dashboard for managing users, records, and academic workflows.

Quantum Key Simulation – Satellite-Based Secure Key Distribution | *Python*

- Built a Python-based **Quantum Key Simulation (QKS)** using the BB84 protocol with a Tkinter GUI. Designed
- **interactive visualizations** for photon transmission, key reconciliation, and error detection.
- Implemented **QBER analysis** to detect eavesdropping and trigger secure key regeneration.
- Developed a **multi-tab GUI** showcasing simulation results, communication visualization, and system logs.

TECHNICAL SKILLS

Languages: JavaScript, Python, SQL, C++

Frontend: HTML, CSS, Tailwind CSS, Bootstrap, React.js

Backend: Node.js, Express.js, JWT

Database: MongoDB, MySQL, PostgreSQL, Firebase

AI/ML: NumPy, Pandas, OpenCV, Scikit-learn, Computer Vision, Data Visualization, Supervised & Unsupervised Learning

Tools: VS Code, Git, GitHub, Firebase, RESTful APIs

Concepts: OOP, DSA

CERTIFICATES & ACHIEVEMENTS

- **Industrial Training certification** – Gained practical exposure to coding standards, scalable solutions, and industry workflows.
- **Internship certification**– Enhanced problem-solving and project execution abilities in a professional environment.
- **NCC 'B' Certificate** – Demonstrated leadership, discipline, teamwork, and organizational skills through National Cadet Corps training.